7.3 - Kruskal's Algorithm

Problem Solving, Math 1000

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Question

Are there algorithms for finding minimum spanning trees? Are some efficient? Are some optimal? Can any algorithm which solves this problem be both? Is the problem of finding minimum spanning trees really all that different from the problem of finding maximum spanning trees?

Algorithm (Kruskal's Algo)

Kruskal's algorithm is an efficient and optimal algorithm for finding minimum spanning trees in a network. The key notion behind Kruskal's algorithm is that we may essentially "build" a minimum spanning tree one edge at a time by choosing the cheapest available in the graph edge at each step of the process. Formally, the algorithm proceeds as follows:

1. Pick the cheapest edge available and mark it (in the case of a tie, pick one at random)
2. Pick the next cheapest edge available and mark it (treating ties similarly once again)
3. Continue picking and marking the cheapest unmarked edge, as long as it does not form a circuit in relation to your previously marked edges. In a network with n vertices, this terminates after $n-1$ steps.

Example

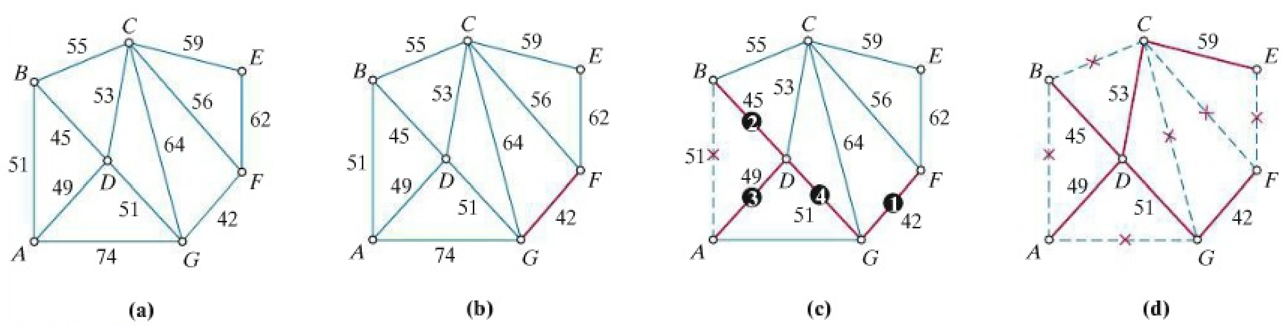


FIGURE 7-17 (a) The original network. (b) Start. (c) First four steps. (d) The MST.

Observe

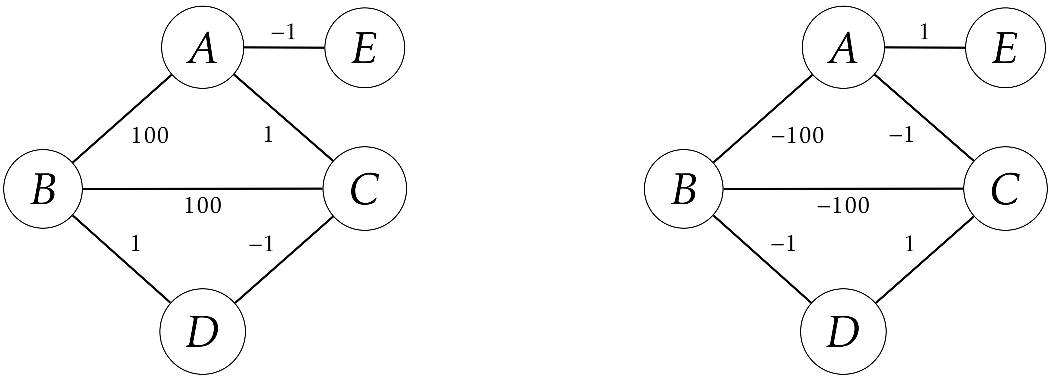
Any algorithm that can find an MST can also find a MaxST – all we must do is change the sign on each weight (i.e. make each positive weight negative and each negative weight positive). This then makes the largest weights the smallest by making them the most negative, and an algorithm like Kruskal's completes the problem.

Definition

For a given weighted network, the network formed by inverting the sign of each weight is called the negative network.

Example

The network on the right is the negative network of the network on the left



Theorem

The MaxST of a network is a MST of the negative network.